MATTHEW C. HEATLEY

[redacted]
https://github.com/sixfourscientific
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OVERVIEW

Bioinformatics Developer & Data Engineer with 10+ years' experience across research, industry & consultancy, bringing cross-disciplinary domain expertise at the cutting-edge of genomics, & backed by a strong record of implementing digital infrastructure across HPCC/cloud environments to tackle real-world big data projects.

SKILLS & KNOWLEDGE

Computational Skills

- **Programming languages**: Python, R, Groovy, SQL, Bash
- **HPCC & cloud systems**: Linux (SLURM, SGE), AWS, GCP
- **Workflow infrastructure**: Docker, Apptainer, Nextflow
- **Data analysis/visualisation**: Pandas, DuckDB, Plotly/Dash
- **Microsoft Office**: Word, PowerPoint, Excel, Power BI

Industry Knowledge & Experience

- **Software development**: Agile, Git, Azure DevOps, Jira
- **Data science, statistics & experimental design**
- **6+ years' lab & applied research experience**
- **Expert witness testimony**

WORK & PLACEMENTS

Jan 2026 - Present | **Bioinformatics Developer & Data Engineer** | **SixFour Scientific**
(Self-Employed) | **Dundee, UK**

As an independent contractor specialising in supporting the life science sector, I construct bespoke computational workflows tailored to project objectives, leveraging best practices & a custom automated build system to substantially reduce development time.

Apr 2023 - Nov 2025 | **Bioinformatician - Genomic Applications** | **Oxford Nanopore Technologies**
Oxford, UK

At the interface of R&D & customer support, I benchmarked a proprietary technology platform & advised on strategy within assigned market segments by developing pipelines to process datasets, evaluating key metrics & presenting my recommendations to stakeholders.

Oct 2021 - Apr 2023 | **Digital Research Scientist - Bioinformatics** | **University of Nottingham**
Nottingham, UK

At the institution's core analytics facility, I provided research groups with dedicated support in all aspects of data science – from advising on study design & developing informatics software to implementing machine learning approaches & wrangling complex datasets.

Oct 2019 - Oct 2021 | **Research Associate – Population Genetics** | **University of Nottingham**
(Fixed-Term) | **Nottingham, UK**

Within an international research consortium, I performed next-generation sequencing & oversaw the population genetics analysis.

Mar 2014 - Sep 2014 | **Research Technician** | **ADAS**
(Fixed-Term) | **York, UK**

At a research station, I provided technical & scientific support across a range of agri-environmental research programmes.

May 2011 - Aug 2013 | **Consultant Scientist** | **CWA International**
London, UK & Worldwide

At a specialist consultancy firm, I undertook forensic analysis on behalf of the legal & insurance sectors. As part of cases, I managed client enquiries, led on-site incident investigations overseas, commissioned analyses & delivered tailored expert advice for use in litigation.

EDUCATION & QUALIFICATIONS

Sep 2015 - Sep 2019 | **PhD** | **University of Nottingham**
BBSRC Doctoral Training Programme (Awarded)

Within the biotechnology research theme, I applied molecular, cytogenetic, cellular & phenotypic techniques informed by phylogenetics & comparative genomics to investigate gene function, with placements at Stanford University, Sheffield University & NIAB/EMR.

Sep 2014 - Aug 2015 | **MSc** | **University of Nottingham**
Biotechnology (Pass with Distinction)

I specialised in plant biotechnology, focusing on genetics, breeding systems, agronomy, bioinformatics & statistics/experimental design.

Sep 2006 - Jun 2009 | **BSc** | **University of Nottingham**
Biology (First-Class Honours)

I opted mostly for subjects in genetics & biotechnology, with a final year lab-based project involving a variety of molecular techniques.

BIOINFORMATICS EXPERIENCE

(Developer)*

Databases	Refgen/Genbank · NCBI-Datasets · Entrez-Direct · SRA-tools · GTDB · Silva · GreenGenes · RDP · Ensembl/Gramene · Biomart
General	BWA · STAR · Minimap2 · MUMmer · Picard · Samtools · Bamtools · VCFtools · BEDtools · Mosdepth · GFFtools · AGAT · Gffread · htseq-count · GFAtools · Guppy · Dorado · Jellyfish · FastK · KMC · Yak · Khmer
Quality Control	FastQC · MultiQC · Nanoplot · NanoOK · Filtlong · MiniScrub · Yacrd · HyPo · Medaka · Fastp · Trimmomatic · Trim Galore · Cutadapt · FASTX-Toolkit · Seqtk · SeqKit · GenomeScope · LocoGSE · Smudgeplot · nQuire · ploidyfrost · QUAST · BUSCO · CheckM2 · Assembly-stats · Inspector · Asset · LAI · Merqury
Variant Discovery	GATK · FreeBayes · Octopus · Paftools · Sniffles · Assemblytics · Minigraph · WhatsHap · IGV · PIMMS*
RNA-Seq	Slamdunk · DEseq2 · Stringtie
Population Genetics	Adegenet · Plink · FastStructure · Scantools*
Genome Assembly	Flye · Canu · Hifiasm · Verkko · NextDenovo · PECAT · NECAT · Meta/NanoMDBG · MyloASM · Trinity · Tellysis · Supernova · Tigmint · Purge_haplotigs · Purge_dups
Genome Annotation	RepeatModeler · EDTA · LTR_retriever · LTR_finder · LTR_harvest · RepeatMasker · DustMasker · Red · Tantan · Augustus · SNAP · MAKER · BRAKER · Prodigal · MetaEUK · RGI · AMRFinder+
Comparative Genomics	SynMap · Circos · Icarus · Ribbon · SplitThreader · Dot · D-Genies
Sequence Analysis	Clustal · PRANK · MAFFT · MUSCLE · T-Coffee · PAL2NAL · Gblocks · Zorro · MeShClust · VSearch · SeaView · BadRead · NanoSim
Phylogenetics	PAML · OrthoFinder · PhyML · iTOL · kWIP
Taxonomy	BLAST · Kraken · Bracken · Centrifuge · Clark · Kaiju · Metabuli · Taxor · Ganon · KMCP · Sylph · Melon · MetaPhlAn · mOTUs · FastANI · Taxpasta · Amber/Opal · Qiime2 · Dada2 · Deblur · Emu · GTDB-Tk · Krona

SCIENTIFIC PUBLICATIONS

Smith, K.E., Cowan, L., Flis, P., Moore, C., Heatley, M., Robles-Zazueta, C.A., Lee, A. & Yant, L. (2024) An ecological, phenotypic and genomic survey of duckweeds with their associated aquatic environments in the United Kingdom. *AoB PLANTS* 17(3); plaf018.

Bellows, E., Heatley, M., Shah, N., Archer, N., Giles, T. & Fray, R. (2024) Comparative transcriptome reprogramming in oak galls containing asexual or sexual generations of gall wasps. *Plant Biology* 26(5); 798-810.

Velasco, A.D.R., Parsons, A.L., Heatley, M.C., Martin, A.R.G., Smart, A.D., Shah, N. & Jopling, C.L. (2024) MicroRNA biogenesis is broadly disrupted by inhibition of the splicing factor SF3B1. *Nucleic Acids Research* gkae505; 1-20.

Smith, K.E., Cowan, L., Taylor, B., McAusland, L., Heatley, M., Yant., L. & Murchie, E. (2024) Physiological adaptation to irradiance in duckweeds is species and accession specific and depends on light habitat niche. *Journal of Experimental Botany* erad499; 1-18.

Yusuf, L., Heatley, M., Palmer, J.P.G., Barton, H.J., Cooney, C.R. & Gossmann, T.I. (2020) Noncoding regions underpin avian bill shape diversification at macroevolutionary scales. *Genome Research* 30(4); 553-565.

Hubbart, S., Smillie, I.R.A., Heatley, M., Swarup, R., Foo, C.C., Zhao, L. & Murchie, E.H. (2018) Enhanced thylakoid photoprotection can increase yield & canopy radiation use efficiency in rice. *Nature Communications Biology* 1(22); 1-12.

REFERENCES

Available on request